

E-Commerce Management System

Module: Order Management Module — End User & Technical Documentation

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GitHub Repo: ansarizeenat/ecommerce-category-management	Live Server (Render): https://ecommerce-category-management.onrender.com
Tech Stack: Java 25, Spring Boot 4.1, JPA, MySQL 8.0, Thymeleaf	Date: September 2026

1. Introduction

The **Order Management Module** facilitates the complete processing and tracking of customer orders within the E-Commerce platform. It enables customers to place orders by selecting products and adding them to a shopping cart with automated calculations for subtotal, taxes, and shipping fees. It provides administrators with an interactive Order Dashboard to view, filter, update statuses (Pending, Shipped, Delivered), and execute order cancellations through a non-destructive soft-delete mechanism that automatically restocks product inventory.

2. Database Design & Specification

The database architecture strictly adheres to the Itvedant requirements with the orders table and associated entities:

Table 1: orders

Column	Data Type	Key / Constraint	Description
id	BIGINT	PK, Auto-Increment	Unique identifier for each order
user_id	BIGINT	FK (users.id)	Foreign key referencing users table (customer)
customer_name	VARCHAR(100)	NOT NULL	Customer name for direct dashboard display
total_amount	DECIMAL(10,2)	NOT NULL	Total cost of order (Subtotal + Tax + Shipping)
subtotal	DECIMAL(10,2)	NOT NULL	Subtotal of ordered products
tax_amount	DECIMAL(10,2)	NOT NULL	Calculated tax (5% GST/VAT)
shipping_cost	DECIMAL(10,2)	NOT NULL	Calculated shipping fee (\$0 if >= \$100, else \$10)
order_status	VARCHAR(50)	NOT NULL	Status: Pending, Shipped, Delivered, Cancelled
shipping_address	VARCHAR(300)	NOT NULL	Shipping destination address
created_at	DATETIME	AUTO-POPULATED	Timestamp when order was placed
updated_at	DATETIME	AUTO-UPDATED	Timestamp when status was last modified
status	BOOLEAN	NOT NULL, DEFAULT TRUE	True for active orders, False for cancelled (soft delete)

Table 2: order_items (Line Items)

Column	Data Type	Key / Constraint	Description
id	BIGINT	PK, Auto-Increment	Unique line item ID
order_id	BIGINT	FK (orders.id)	Foreign key referencing parent order
product_id	BIGINT	FK (products.id)	Foreign key referencing catalog product
product_name	VARCHAR(255)	NOT NULL	Product snapshot name at time of order
quantity	INT	NOT NULL	Quantity purchased
price	DECIMAL(10,2)	NOT NULL	Unit price at time of purchase
item_total	DECIMAL(10,2)	NOT NULL	Line total: quantity * price

3. Core Functionalities & Business Logic

- 1. Place an Order:** Customers add products to the shopping cart, specify quantities, and provide shipping details. The system calculates: Subtotal = Sum(Item Price * Quantity), Tax = Subtotal * 5%, and Shipping Cost = Free (\$0) if Subtotal >= \$100 else \$10.00. On order placement, product stock is decremented.
- 2. Order Dashboard:** Admins view all orders in reverse chronological order. Quick status filter tabs (All, Pending, Shipped, Delivered, Cancelled) provide fast triage. Each record displays Order ID, Customer Name, Shipping Address, Items Count, Total Amount, Order Status badge, Active Status, and action buttons.
- 3. Update Order Status:** Admins can progress order lifecycle (Pending -> Shipped -> Delivered). Last updated timestamp is refreshed automatically.
- 4. Order Cancellation (Soft Delete):** Customers and admins can cancel orders only if not yet shipped (Pending). Cancellation performs a soft delete: orders.status is set to false, order_status is set to Cancelled, and reserved inventory is automatically restored back to the catalog.

4. REST API Endpoint Specifications

Method	Endpoint	Description	Sample Payload / Response
POST	/api/orders	Place a new order with cart items	{"customerName": "Zeenat", "shippingAddress": "Mumbai", "items": [{"productId": 1, "quantity": 2}]}
GET	/api/orders	Retrieve all orders (supports ?status=Pending)	Returns JSON list of orders sorted newest first
GET	/api/orders/{id}	Fetch single order with full item details	Returns order object with line items array
PUT	/api/orders/{id}/status	Update order status	{"orderStatus": "Shipped"}
DELETE / PUT	/api/orders/{id} /api/orders/{id}/cancel	Soft-delete cancel order (Pending only) & restock stock	Sets status=false, orderStatus='Cancelled'

5. Automated Test Suite & Verification

Automated unit and integration tests were implemented in
src/test/java/com/ecommerce/ecommerce/OrderServiceTest.java:

- **testPlaceOrder_Success_AndCalculation:** Verified Subtotal, 5% Tax, and Free Shipping calculations.
- **testPlaceOrder_WithShippingCost:** Verified \$10 shipping fee applied for orders below \$100.
- **testPlaceOrder_InsufficientInventoryThrowsException:** Verified inventory boundary validation.
- **testUpdateOrderStatus:** Verified admin status transition.
- **testCancelOrder_SoftDeleteAndInventoryRestock:** Verified soft delete (status=false) and restock.
- **testCancelOrder_AlreadyShippedFails:** Verified cancellation is rejected for shipped orders.

Result: [INFO] Tests run: 7, Failures: 0, Errors: 0, Skipped: 0 — BUILD SUCCESS

6. Assignment Submission Summary

Item	Details & Links
1. GitHub Repository	https://github.com/ansarizeenat/ecommerce-category-management
2. Free Cloud Hosted URL	https://ecommerce-category-management.onrender.com
3. End User Documentation	Comprehensive guide generated in PDF format (attached document).
4. Modules Completed	1. Category Management (Complete) 2. Product Management (Complete) 3. Order Management (Complete)